

Mohit Kumar

SOFTWARE DEVELOPER & AI RESEARCHER

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CORE EXPERTISE

DICOMweb

DICOM/HL7

PACS

MONAI

PyTorch

3D U-Net

Swin UNETR

ITK/VTK

FastAPI

Docker

GitHub Actions

HL7 FHIR

Radiomics

Ollama

LangChain

RAG

React

Plotly

Netlify

Gymnasium

EDUCATION

AI for Clinicians

NPTEL (IIT Madras)

Deep Learning in Healthcare

Stanford / DeepLearning.AI

Healthcare AI Specialization

Medical Imaging & Clinical Validation

B.E. / B.Tech

CS / ECE — AI/ML & Medical Imaging

PUBLICATIONS

AI in Interventional Procedures

Narrative Review · Preprint 2026

Photon-Counting CT Review

Technical Review · Published Aug 2026

DICOMweb AI Gateway

Technical Article · 5 Figures · E1–E4 Protocol

LINKS

GitHub

greyentity101

Live Site

alfredai-site · keenkasha.netlify.app

Key Repos

dicomweb-ai-gateway, alfred-rag, photon-counting-ct

PROFESSIONAL SUMMARY

Software Developer and AI Researcher specializing at the intersection of **medical imaging** and **deep learning**. Experienced in building end-to-end healthcare AI pipelines—from DICOM/HL7 ingestion and preprocessing to model inference, evaluation, and deployment. Strong focus on **local-first**, **privacy-preserving inference** and **clinical workflow automation**. Proven ability to ship production-grade tooling, academic publications, and autonomous agent systems from concept to live deployment.

SELECTED PROJECTS

DICOMweb AI Gateway

Lead Developer & Researcher · github.com/greyentity101/dicomweb-ai-gateway

- ▶ Architected a **DICOMweb-native AI inference gateway** enabling seamless deployment of segmentation and detection models on radiological imaging data.
- ▶ Designed and implemented the **E1-E4 evaluation protocol**: latency benchmarking (p95), 3D segmentation accuracy (Dice/HD95), anatomical grounding, and end-to-end round-trip validation.
- ▶ Built reproducible synthetic dataset generators for chest CT and brain MR with known ground-truth masks, enabling offline model validation without PHI.
- ▶ Published technical article with **5 publication-grade figures**; repository includes full evaluation suite, metrics reporters, and CI-tested smoke tests (21/21 passing).

Alfred AI Site — Research Portfolio

Founder & Full-Stack Engineer · greyentity101.github.io/alfredai-site

- ▶ Built and deployed a **privacy-first, offline-capable research portfolio** with real-time search, citation export, and theme persistence.
- ▶ Engineered a publication management pipeline that syncs assets (papers, abstracts, figures) from research repositories to the live site via automated GitHub Actions.
- ▶ Implemented search/filter UI with **localStorage persistence**, animated mobile navigation, and glass-morphism UI components.
- ▶ Achieved **HTTP 200, sub-second load times**, and offline service-worker caching for uninterrupted access.

Alfred RAG — Local Clinical Document Retrieval

Developer · github.com/greyentity101/alfred-rag

- ▶ Built a **local, model-agnostic RAG pipeline** for semantic retrieval over clinical and technical documents.
- ▶ Implemented on-device embeddings via Ollama `omic-embed-text` — no API keys, no cloud egress.
- ▶ Designed smart 600-token chunking with overlap and sentence-boundary breaking, stored in portable JSON indexes.
- ▶ Supports `.md`, `.txt`, `.csv`, `.pdf`, `.docx`, `.json` with three commands: `index`, `search`, `status`.

Photon-Counting CT Review

Author & Researcher · github.com/greyentity101/photon-counting-ct

- ▶ Authored a **400-line technical review** on photon-counting CT detectors (CdTe/CZT and deep silicon).
- ▶ Covered physics, materials science, readout electronics, vendor landscape, clinical evidence, and PACS integration challenges.
- ▶ Generated **8 analytical figures** including mass attenuation curves, detector physics, count-rate limitations, and spectral material decomposition.
- ▶ Published August 2026; available at keenkasha.netlify.app.

Last updated: August 2026

AI in Interventional Procedures

Author & Researcher · github.com/greyentity101/AI-Interventional-Procedures

- ▶ Authored a **comprehensive 469-line narrative review** on AI in interventional procedures across cardiology, radiology, oncology, neurointervention, and endovascular surgery.
- ▶ Examined ML/DL mechanisms for procedural planning, risk prediction, imaging guidance, robotics, and digital twins.
- ▶ Synthesized clinical applications, validation strategies, regulatory considerations, and cost-effectiveness data.
- ▶ Preprint published with ORCID 0009-0006-4312-403X.

Neural Phase Retrieval (Active)

Researcher & Engineer · [Fortis Hospitals / Independent](#)

- ▶ Developing a **physics-informed U-Net (PyTorch)** that compensates for hardware micro-vibrations in X-ray dark-field radiography.
- ▶ Deployed as a **DICOM C-STORE SCP** with edge inference routing diagnostic maps to PACS in under 50 ms.
- ▶ Bridges clinical imaging infrastructure with real-time AI correction at the acquisition console.

SWE-RL-Envs

Developer · github.com/greyentity101/swe-rl-envs

- ▶ Built **Gymnasium environments** for software-engineering workflows — deterministic test-driven debugging and CI-style tasks.
- ▶ Designed SweDebug-v0 where RL agents propose (`file`, `fix` `strategy`) actions and receive reward signals based on test outcomes.
- ▶ Enables fast, deterministic benchmarking of code agents before graduating to real SWE benchmarks.

PUBLICATIONS

- "The Predictive Role of AI in Reducing Complications in Interventional Procedures" — Narrative review, 469 lines, covering ML/DL across interventional cardiology, radiology, oncology, neurointervention, and endovascular surgery. Preprint 2026. ORCID: 0009-0006-4312-403X.
- "Photon-Counting CT Detectors: Silicon, CdTe, and the Road to Clinical Spectral Imaging" — Technical review, 400 lines, 8 analytical figures, covering semiconductor physics, vendor landscape, and PACS integration. Published August 2026.
- "DICOMweb AI Gateway: A Vendor-Neutral Framework for Deploying Medical Imaging AI Models" — Technical article, 2,367 words, 5 publication-grade figures, full E1–E4 evaluation protocol. Repository includes reproducible evaluation suite.

EDUCATION & CERTIFICATIONS

- ▶ **AI for Clinicians** — NPTEL (IIT Madras)
- ▶ **Deep Learning in Healthcare** — Stanford / DeepLearning.AI
- ▶ **Healthcare AI Specialization** — Medical image analysis, regulatory considerations, clinical validation
- ▶ **Autonomous Agent Systems** — Local-first AI orchestration, multi-model routing, Hermes agent architecture
- ▶ **B.E. / B.Tech** — CS / ECE with specialization in AI/ML and Medical Imaging

OPEN SOURCE & COMMUNITY

- ▶ **GitHub:** [greyentity101](#) — Active contributor to medical AI tooling, local-first agents, and healthcare open-source projects.
- ▶ **Repositories:** [dicomweb-ai-gateway](#), [alfredai-site](#), [alfred-rag](#), [photon-counting-ct](#), [AI-Interventional-Procedures](#), [swe-rl-envs](#).
- ▶ **Interests:** Privacy-preserving medical AI, local LLM deployment, DICOM standardization, clinical workflow automation, imaging informatics.